NEUTRON THERAPY FACILITIES: STATUS

STATUS	REACTOR	D-T	CYCLOTRON	LINAC	TOTAL
Operational	1	0	6	1	6-7
Terminated	2	7	20	1	30
Unknown	0	1	2	0	3
	1	1 (?)	23		33
					41

(Update: Jan. 2011 by FM Wagner)

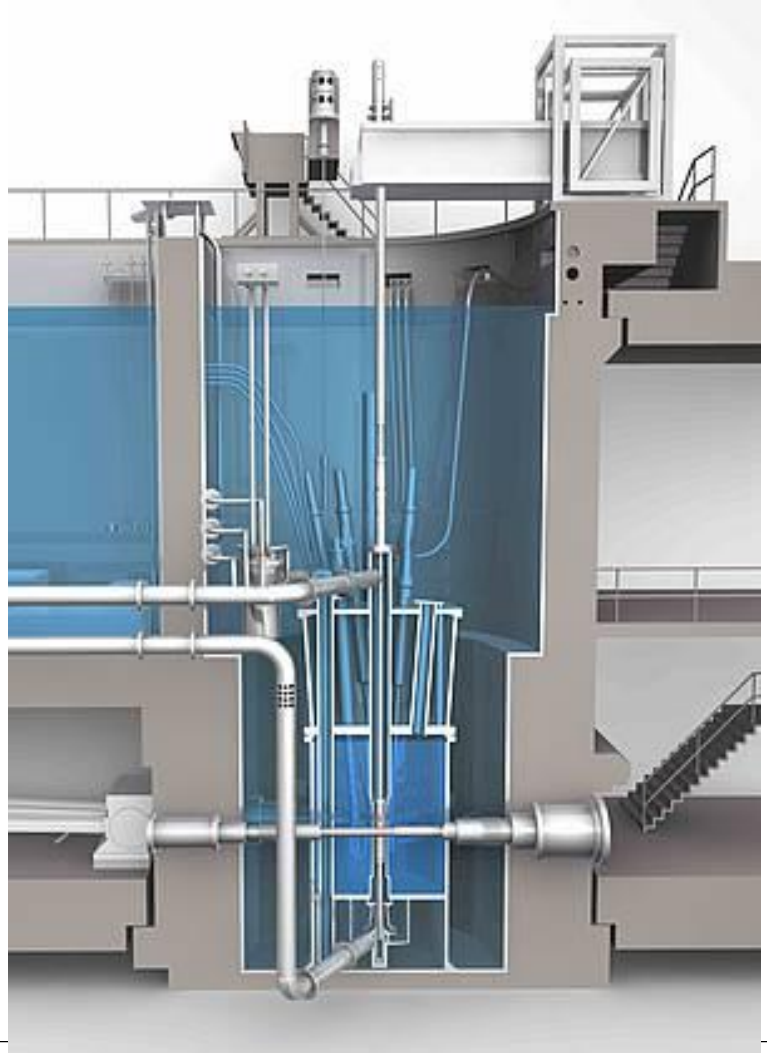
FAST NEUTRON THERAPY FACILITIES (8)

Oct. 2010

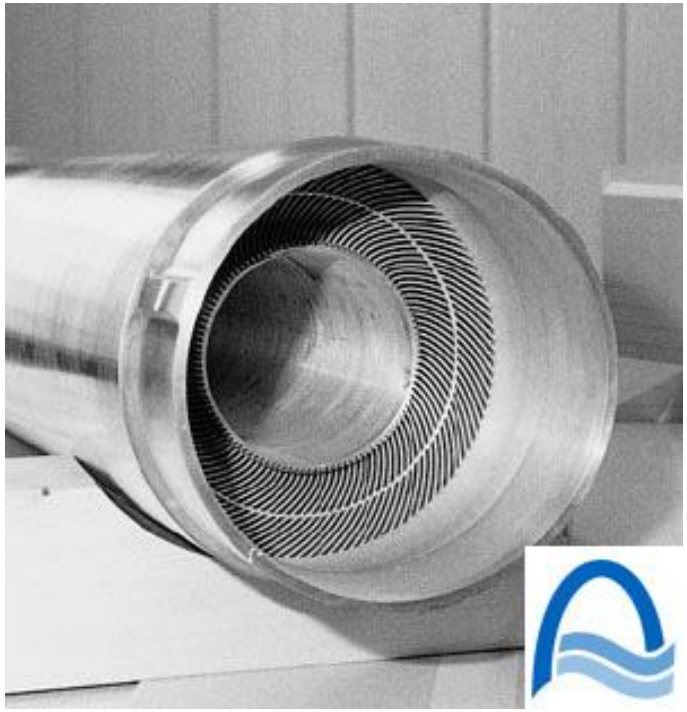
LOCATION	COUNTRY	SOURCE REACTION	SAD (cm)	BEAM DIRECTION	COLLIMATOR TYPE	FIRST PATIENTS	(March 2006)
Munich FRM I+II						1985/2007	715-1047
~~Garching (2)~~	Germany	Reactor	620	Horizontal	Multileaf	2006	
~~Essen~~ ▪	Germany	d(14.3) + Be	125	Isocentric	Inserts	1978	763
Orleans ▪	France	p(34) + Be	169	Vertical	Inserts	1981	2103
Detroit MI ▪	USA	d(48.5) + Be	183	Isocentric cyclotron	Multirod	1990	2100
Seattle WA (2) ▪	USA	p(50.5) + Be	150	Isocentric Horizontal	Multileaf Inserts	1984	2489
~~Seoul~~ ▪°	South Korea	p(50.5) + Be	150	Isocentric	Variable jaws	1986	310
Batavia IL •	USA	p(66) + Be	190	Horizontal	Inserts	1976	3309
Somerset West ▪	South Africa	p(66) + Be	150	Isocentric	Variable jaws + Multiblade trimmer	1988	1396
Tomsk	**Russia**	**p+Be**					**1200**
Snezhinsk°	**Russia**	**d-T**		**hor.**	**Inserts**	**1999**	**990**

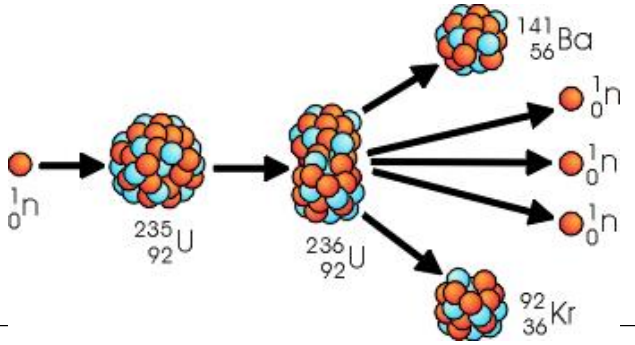
Forschungsneutronenquelle Heinz Maier-Leibnitz (FRM II)
Technische Universität München, Garching

Reaktorbecken

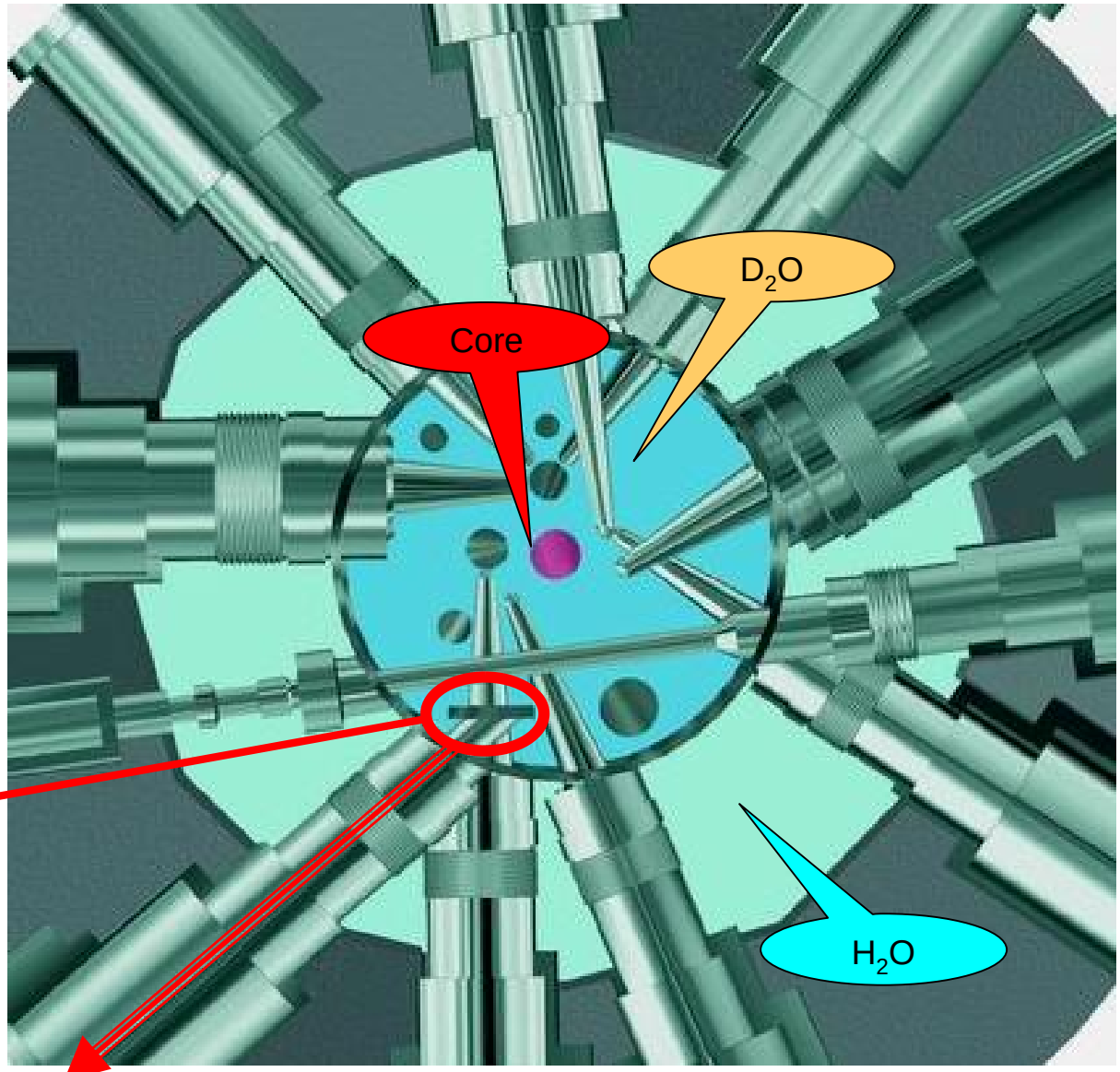


Brennelement

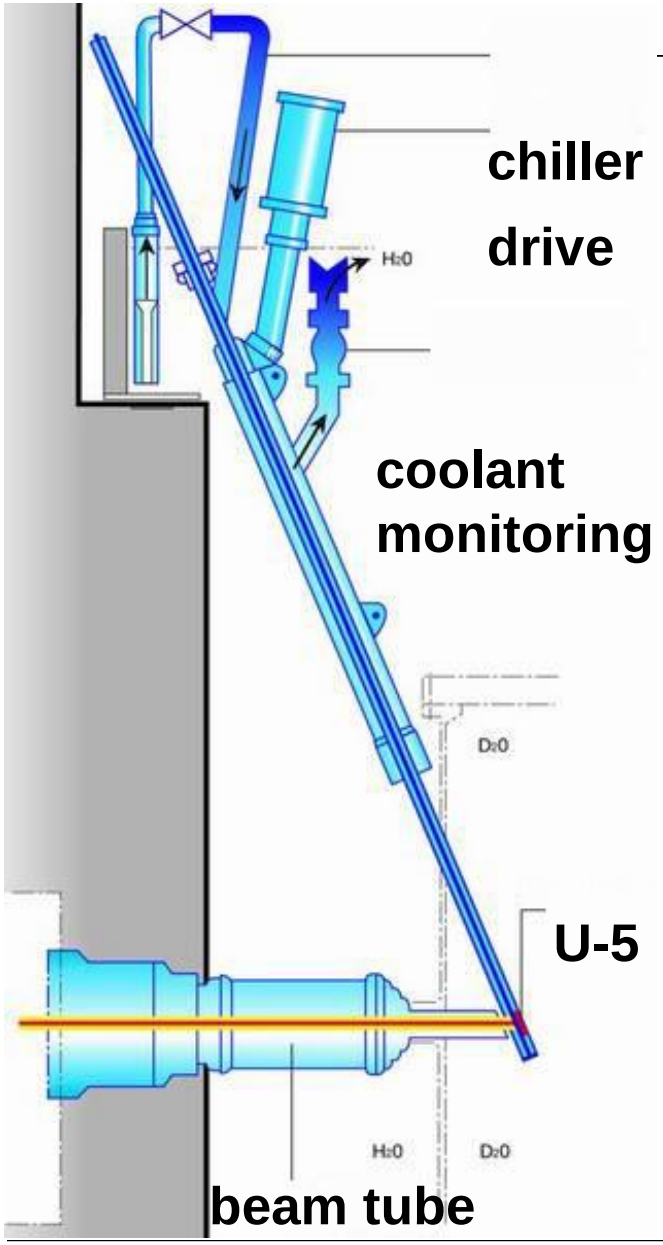


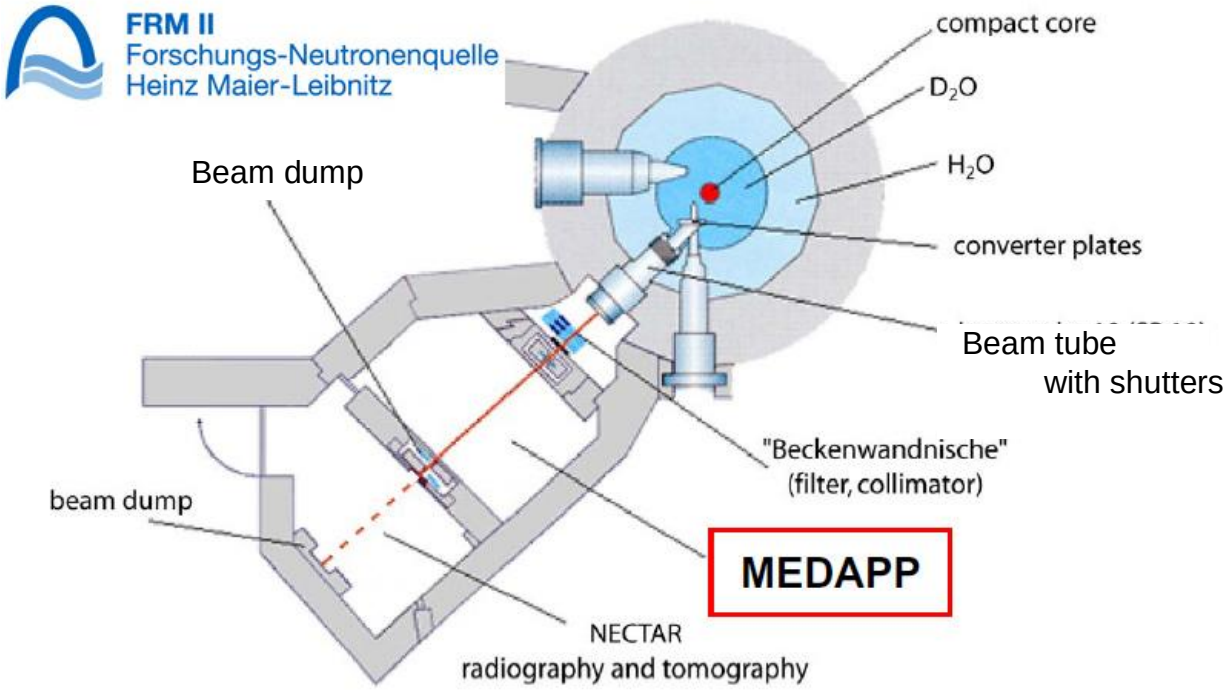


Reactorpool

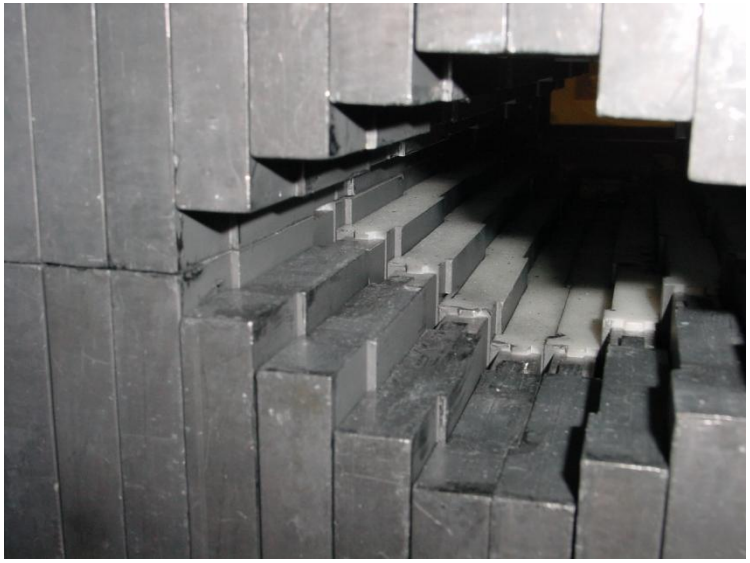


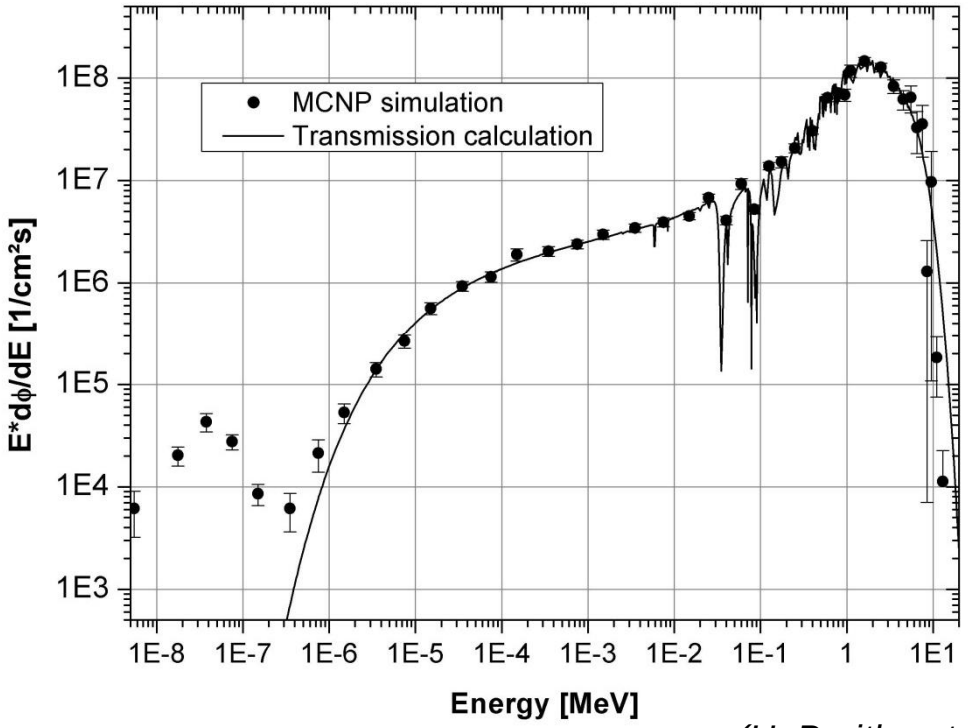
Thermal-to-Fast Neutron Converter



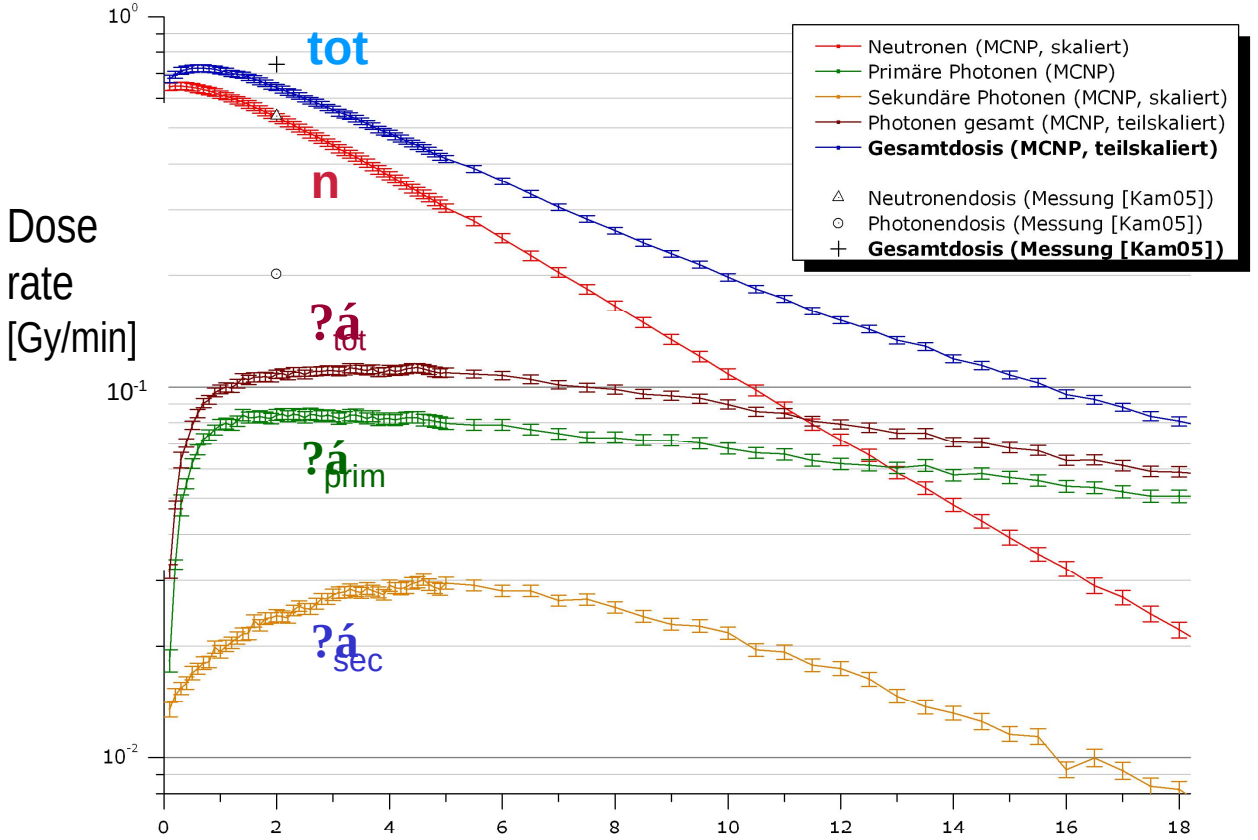


Bestrahlungsraum, Kollimator

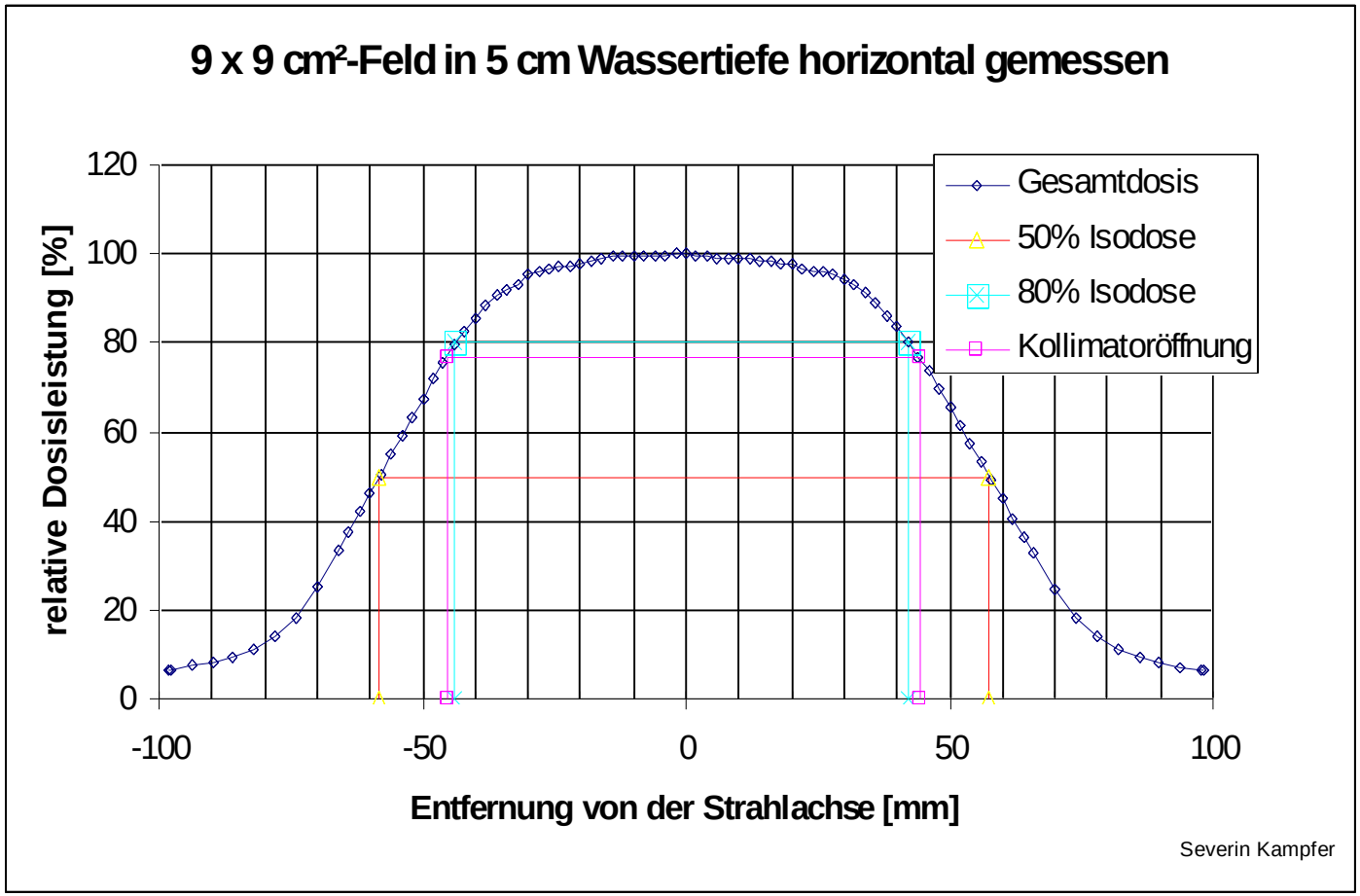




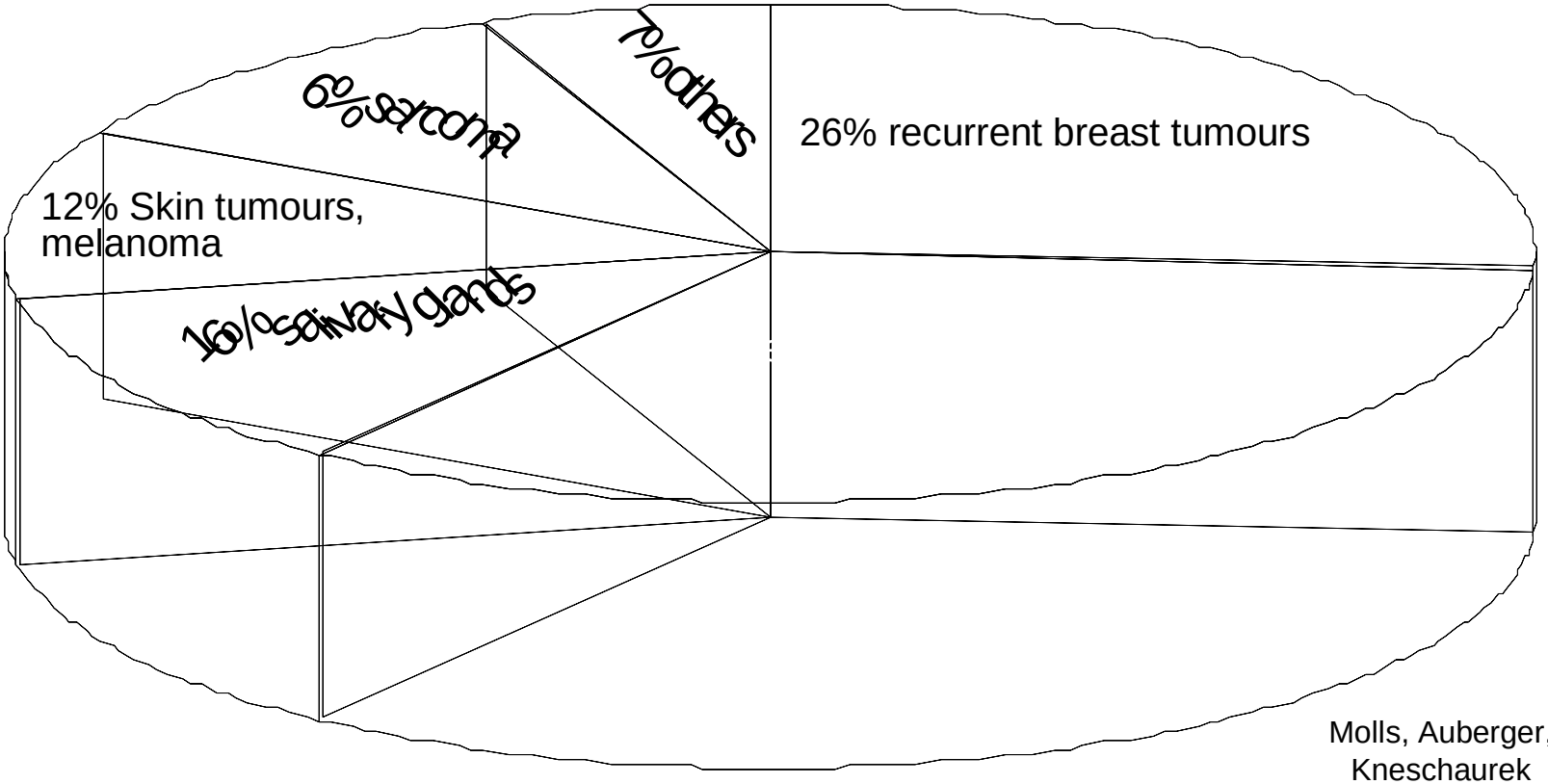
(H. Breitkreutz)



(H. Breitkreutz)

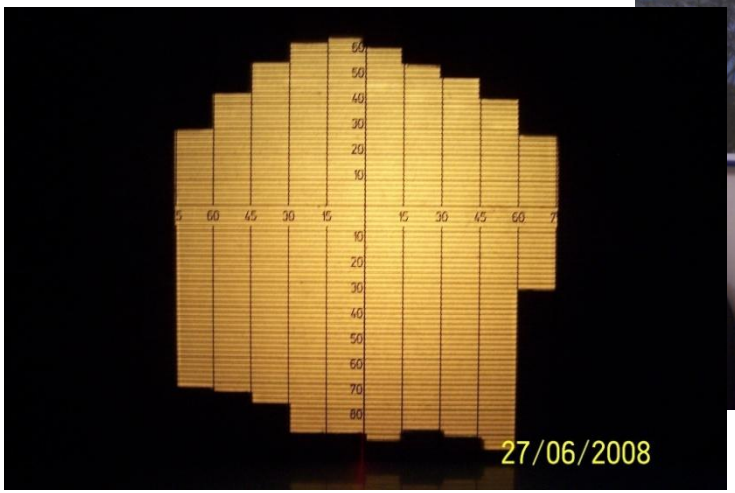

9 x 9 cm²-Feld in 5 cm Wassertiefe horizontal gemessen
relative Dosisleistung [%]
120
100
80
60
40
20
0
-100
-50
0
50
100
Entfernung von der Strahlachse [mm]
Gesamtdosis
50% Isodose
80% Isodose
Kollimatoröffnung
Severin Kampfer

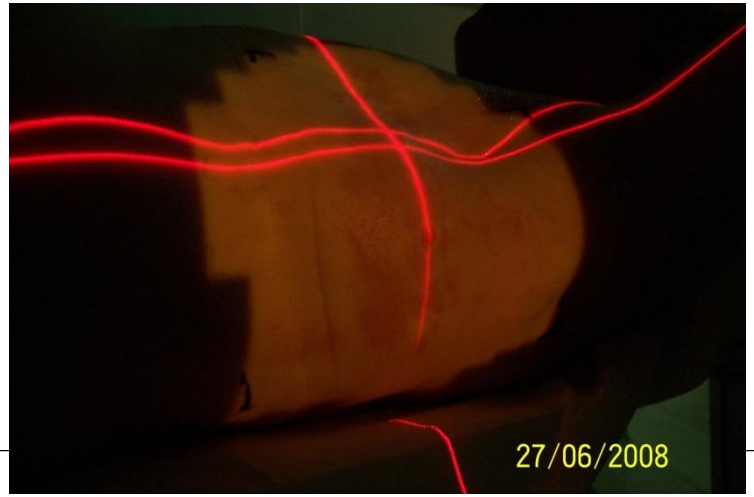
Indikationen zur Bestrahlung mit Spaltneutronen

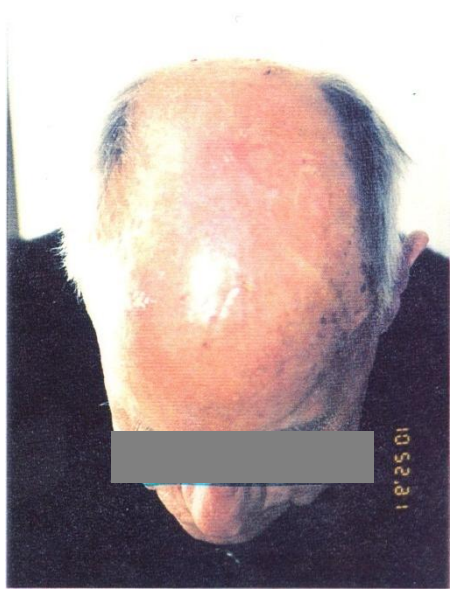


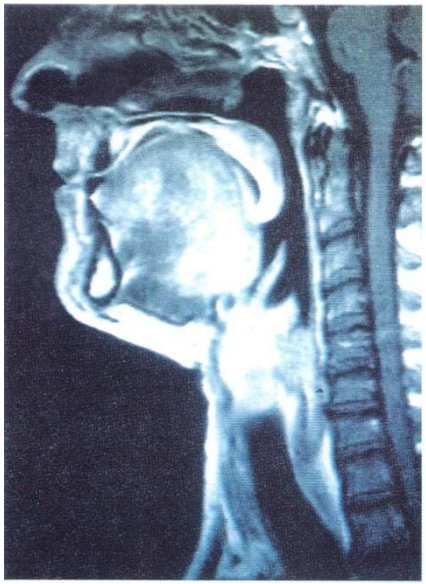
Molls, Auberger, Kneschaurek

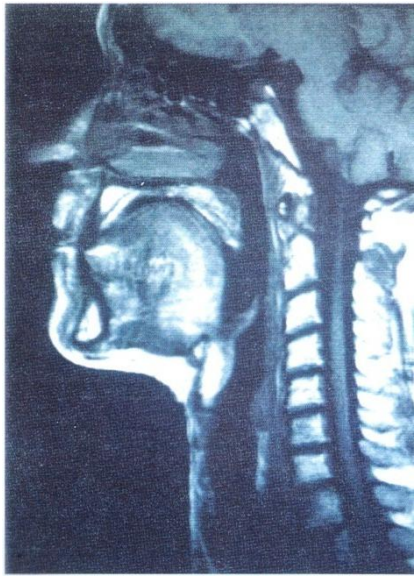
Brustwand-Metastasen eines Mamma-Ca: Verlauf











FAST NEUTRON THERAPY

Reflections - .

- ± 27 000 patients treated to date (estimated)
- Efficacy for several tumor types has been well established in randomized and other clinical trials
- But, many facilities were closed because of poor clinical results
  - side effects due to poor physical beam characteristics
    - poor penetration (low energy)
    - few had gantries
    - few had flexible beam shaping
  - unethical to treat patients if x-ray beams had similar characteristics
    - better treatments in photon arms of trials?
- Neutrons got a poor reputation, which has never been regained
- Viewed as outdated modality (~ radium)
- The advantages of neutrons are seriously underestimated